General Instruction

1. Student Instruction

Basic Rules

1. Each team must consist of 1–4 students.
2. Every team must submit via GitHub Repository Link
    - Working Project (Frontend + Backend)
    - Final Presentation (PPT and Demo)
3. Project must be:
    - Original
    - Developed during the hackathon timeline
4. AI tools (ChatGPT, Copilot, etc.) are allowed for guidance, but:
    - You must understand your code.
    - You must explain your logic during judging.

Timeline Discipline

- No late submissions.
- Final Git commit time will be verified.
- Last-minute ZIP submissions without proper repository history will be rejected.

Project Expectations

Your project must include:

- Working UI
- Proper Backend/API
- Database integration
- Deployment (recommended and Preferable)

2. Git Repository & Team Development Rules

Mandatory Requirements

1. Repository must be created by the team leader within 24 hours of releasing the problem statement.
2. Repository name format: NavKalpana-TeamCode
   Example: NavKalpana-RICR-NK-0001
3. All team members must:
    - Must be the contributor in repo
    - Use individual GitHub accounts
    - Contribute via commits
    - Avoid single-person commits for team projects

Commit Rules

- Minimum 10 meaningful commits.
- Avoid a single bulk commit at the end.
- Commit messages must be clear.
    - Good Commit Example:
        - Added login authentication
        - Integrated payment API
        - Fixed cart validation bug

- Bad Commit Example:
  - Update
  - Final
  - Added
  - done

Bad Commit Messages will lead to negative marking

Branching Rule (Recommended)

- main – Final stable version
- All the Members must Create Separate Branches which further merges to the Main Branch
  - Example:
    - feature/login
    - feature/register
    - bug/update etc.

3. Recommended Project Folder Structure

Repository Name Example: *NavKalpana-RICR-NK-0001*

Standard Structure

- NavKalpana-TeamCode
  - frontend
  - backend
  - docs
    - problem-statement.pdf
    - architecture-diagram.png
    - api-documentation.md
    - presentation.pptx
  - README.md

4. README.md Must Include

- Project Title
- Team Members & Roles
- Problem Statement
- Tech Stack Used
- Installation Steps
- API Endpoints
- Screenshots
- Future Improvements

5. What Judges Will Verify

Technical Verification

- Git commit history
- Contribution by all members
- Code understanding

- Folder structure clarity
- Backend API functionality
- Database integration
- Error handling

Innovation Check

Judges will evaluate:

- Whether it solves a real problem
- Practical usability
- Scalability potential
- Meaningful use of AI (if included)

Viva Round Questions

Each member may be asked:

- Explain your module.
- Explain database schema.
- Explain API flow.
- Explain deployment steps.
- Explain one technical challenge you faced.

6. Important Warning

Immediate disqualification may occur if:

- Code is copied from the internet without modification
- Purchased templates are used without proper logic implementation
- Repository is created only on the final day
- Only UI is presented without functionality
- Fake data is shown without backend logic (unless clearly defined prototype)
- Only one member answers all questions

7. Bonus Considerations

- Clean UI/UX
- Proper Authentication
- Role-based access control
- AI/ML Integration (if required)
- Deployment (Vercel, Render, AWS, Azure)
- Clean code practices

Final Note

The hackathon is not only about coding. It is about teamwork, structured thinking, problem solving, and clear communication. Build a meaningful solution and be prepared to explain it confidently.

Student Learning & Progress Platform
(Student Module)
Product Overview

The Student Learning & Progress Platform is a secure web-based system designed for registered students.
After login, students will access a Student Dashboard, which acts as the central control panel for:

- Courses
- Assignments
- Quizzes
- Attendance
- Learning Support
- Progress Tracking
- Job & Internship Section
- Alumni Section

Note: Create proper side or top navigation for these sub-pages.

Part 1 – Core Foundation

1. Authentication Module

Objective

Provide secure access to registered students only.

Features

- Email and Password login
- Password hashing (bcrypt recommended)
- JWT-based session handling
- Logout functionality
- Session timeout (approximately 24 hours)

Functional Flow

Login → Validate Credentials → Generate JWT → Redirect to Dashboard

2. Student Dashboard (Landing Page After Login)

This is the first page visible after login.
The UI design is flexible. Students may design the layout as per their preference. However, the following functional components are mandatory.

2.1 Greeting Section

- Time-based greeting
- Student Name

2.2 Academic Score Card

Display overall performance percentage.
Must automatically update whenever there is completion or evaluation in:

- Assignments
- Quizzes
- Courses

2.3 Assignment Summary Card

Display Completed Assignments / Total Assignments
Must update upon:

- New submission
- Evaluation of assignment

2.4 Learning Streak

Display continuous learning days.
A day will be counted if:

- Lesson opened
- Assignment submitted
- Quiz attempted

2.5 Skills Acquired

Display Acquired Skills / Total Skills
Skills linked to module completion.

2.6 Weekly Learning Activity Chart

Display:

- Lessons completed per day
  or
- Time spent per day

2.7 Events Calendar

Display:

- Assignment deadlines
- Quiz dates
- Platform events

Dummy data allowed.

2.8 Leaderboard

Display:

- Student Name
- Score

Dummy data allowed.

3. Courses Module

3.1 My Courses Page

Each course card must show:

- Course Name
- Progress Percentage
- Instructor Name
- Attendance Percentage
- Thumbnail (optional)

Progress Formula
Progress Percentage = Completed Lessons / Total Lessons × 100

Demo Requirement – Course Completion

Include button: “Mark Course as Complete”
This button must:

- Mark all modules completed
- Mark all lessons completed

- Set course progress to 100%
- Update Dashboard statistics

Demo purpose only.

3.2 Course Detail Page

Each module must show:

- Module Name
- Expand/Collapse
- Module Progress Percentage

Each lecture must include:

- Lecture Title
- Difficulty Level
- Video
- Notes
- Quiz
- CodeLab (optional)

Lesson Completion Requirement:

Include button: “Mark Lesson as Complete”

When clicked:

- Lesson marked complete
- Module progress updates
- Course progress updates
- Dashboard score updates
- Learning streak updates

All updates must reflect real-time data.

4. Assignment Module

Each assignment must include:

- Title
- Description
- Deadline

Submission Types:

- File Upload
- Text Input
- External Link

System must store:

- Submission Time
- Late Flag
- Status (Not Submitted, Submitted, Late Submitted, Evaluated)

Display:

- Marks
- Feedback

Dummy evaluation allowed. Submission must be real-time.

5. Quiz Module

Each quiz must include:

- Title
- Duration
- Total Questions

During quiz:

- Countdown timer
- MCQ (single/multiple correct)

After submission:

- Score Percentage
- Correct / Incorrect count
- Optional explanation

Quiz Formula

Score Percentage = Correct Answers / Total Questions × 100

Include 4–5 dummy quizzes. Must update metrics in real-time.

6. Attendance Module

Students must be able to:

- View attendance percentage per course
- View present/absent history
- View monthly summary

Attendance Formula

Attendance Percentage = Present Days / Total Classes × 100

Dummy data allowed.

7. Learning Support Module

Provide:

- Doubt Submission
- Request Backup Class

Students must be able to:

- Select course
- Select topic
- Describe doubt (for doubt submission)
- Upload screenshot/file (optional)

Status tracking optional.